

# Input Form & Map Generator — What Must Change

Goal: one perfect pipeline — **input form** → **construction-usable 2D site plan**. Prepared by Fable (Claude), reviewed with Sol · 25 July 2026 · Audit only, no code changed.

| Input form today | Drawing output today        | Status                                               |
|------------------|-----------------------------|------------------------------------------------------|
| 4 / 10           | 3 / 10 for construction use | <b>Blocked</b> on 4 questions for Shivam (last page) |

## Part 1 — What the input form needs to become better

The core problem: the current form is a **finance form, not a survey form**. Of its 14 fields, only 5 affect the drawing at all. Three *required* fields — budget, land ownership, optimisation priority — have zero effect on geometry, yet the form refuses to draw without a budget. A construction drawing should never be gated behind a budget figure.

### 1.1 — What each current field actually does

| Field                    | Req. | Effect on the drawing                                                      |
|--------------------------|------|----------------------------------------------------------------------------|
| Project name             | Yes  | Title block text and filename only                                         |
| Jurisdiction (6 options) | Yes  | Picks bylaw table: setbacks, FAR/coverage caps, parking rate               |
| Master-plan land use     | Yes  | Permissibility check + title text. Does NOT constrain the layout           |
| Development type         | Yes  | Picks layout generator branch and density caps                             |
| Plot area (acres / sq m) | Yes  | Sole geometry source without dimensions - a rectangle is INVENTED at 1.4:1 |
| Plot width / depth (m)   | Opt  | Both -> exact rectangle. One -> other side derived from area               |
| Abutting road width (m)  | Yes  | High-rise/fire checks only. The road is never drawn                        |
| Sanctioned FAR / FSI     | Opt  | Raises massing ceiling; noted in title block                               |
| GPS pin                  | Opt  | Title-block text only - no geometric effect                                |
| Budget (Rs crore)        | Yes  | NOTHING - feasibility only, yet it gates the drawing                       |
| Land ownership           | Yes  | NOTHING - feasibility cost line only                                       |
| Optimisation priority    | Yes  | Changes the drawing from a marketing preference, not a site fact           |
| Target units             | Opt  | Unit-mix table only, not the drawing                                       |
| Notes                    | Opt  | Stored; no effect                                                          |

### 1.2 — What is missing entirely

A drawing a contractor can set out from starts with a **survey**, and the form cannot describe one.

**Boundary geometry (most critical).** All side lengths PLUS at least one diagonal, or corner coordinates/bearings from a survey - today the plot is always assumed rectangular. North angle / orientation - today north is hardcoded "up". Which edge(s) face a road and the road width per edge - today road position is not even asked. Local units: square yards / gaj, the NCR standard - today only acres and sq m exist.

**Plot identity.** Sector / pocket / plot number, khasra number, scheme or licence number. Today "project name" is free text.

**Site context.** What adjoins each boundary (road, neighbour, park, drain). Existing structures, trees, poles, HT lines, drains. Levels: benchmark and plot level. Easements / right-of-way restrictions.

**Drawing intent.** Purpose: site plan vs authority-submission vs construction setting-out - each has different dimensioning standards. User-specified vs generated building footprint. Required formats (DWG / DXF / PDF).

**The standard to aim at:** the form is perfect when it captures everything found on an accepted Dwarka (DDA) or Gurugram (DTCP) site plan and nothing else - no field the drawing does not use, no drawing element that did not come from a field. Sol/Codex is verifying the official requirements; the two lists merge into one agreed specification.

## Part 2 — Hidden assumptions in the current map generator

Each of these is invisible to the user, and each one on its own makes the output unsafe to build from.

- 1. The plot is a perfect rectangle.** No irregular shapes, bearings, or corner coordinates. If your plot is not rectangular, the drawing is of a different plot.
- 2. North always points up.** Hardcoded. The north arrow does not reflect the real orientation of the land.
- 3. The road is on one assumed edge - and is never drawn.** The generator treats the top edge as frontage internally; the user cannot say which side the road is on.
- 4. Setbacks are drawn but not enforced.** THE MOST SERIOUS DEFECT. Buildings are placed at 0 m from the boundary while rule-screening reports setbacks as passing, because the screen reports a fabricated value (110% of requirement) instead of measuring the geometry. Measured: required 9 m front / 6 m side / 6 m rear; actual building line 0.0 m on all sides; screen says "pass".
- 5. The layout depends on a marketing choice, not the site.** The same plot produces three different drawings from an ROI/balanced/green preference. Internal road widths are derived from plot size, not from any standard or input.
- 6. All geometry snaps to 0.5 m; labels round further.** Fine for a concept sketch; unacceptable for setting out.
- 7. Height is floors x 3.1 m, flat.** No plinth, stilt, or basement input exists anywhere.
- 8. The site is assumed empty and flat.** No existing structures, trees, services, levels, or benchmarks - nothing a contractor actually pegs from.

**What IS trustworthy:** the file container. The DXF/PDF writers are structurally valid, deterministic, true to scale, and honestly labelled (verified by parsing and measuring the files). The problem is not the files - it is the geometry inside them.

## Part 3 — Division of work (Fable ↔ Sol/Codex)

**Fable (Claude):** form experience and integration - survey-grade boundary entry (sides + diagonal, orientation, road edges, gaj/sq yd), geometry pipeline, DXF/PDF writers.

**Sol/Codex:** official DDA/DTCP requirements, the input-to-drawing standard, CAD/MCP workflow research, geometric validation tests, acceptance checklist. Fail-closed gating stays: unverified input is labelled, never assumed.

**Rules:** one shared SitePlanBrief specification agreed before any code; Sol's geometric tests are written BEFORE Fable refactors the layout engine; each reviews the other's work.

## Part 4 — Four questions blocking all work (for Shivam)

|                                  |                                                                                                                                              |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Which area first?             | Dwarka Sub-City (DDA), Dwarka Expressway / Gurugram (DTCP), or both?                                                                         |
| 2. Which drawing?                | Site plan, authority-submission drawing, or construction setting-out drawing?                                                                |
| 3. What documents already exist? | Survey report, sale deed dimensions, sanctioned plan - anything with measured boundaries? One real example (even a photo) sets the standard. |
| 4. Which output formats?         | DWG, DXF, PDF, or all three? True DWG needs a conversion step outside the browser - this answer decides the architecture.                    |

Until these are answered, both agents build nothing.